Web Basics

Python Course · Module 8

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Module 8 — Web Basics

We connect Python skills to the web stack:

We will go in three stages:

1. **Web pages** — HTML, CSS, JavaScript, and how the browser talks to a server
2. **HTTP & APIs** — requests, responses, status codes, and JSON
3. **Python as a web client/server** — using requests and serving simple pages

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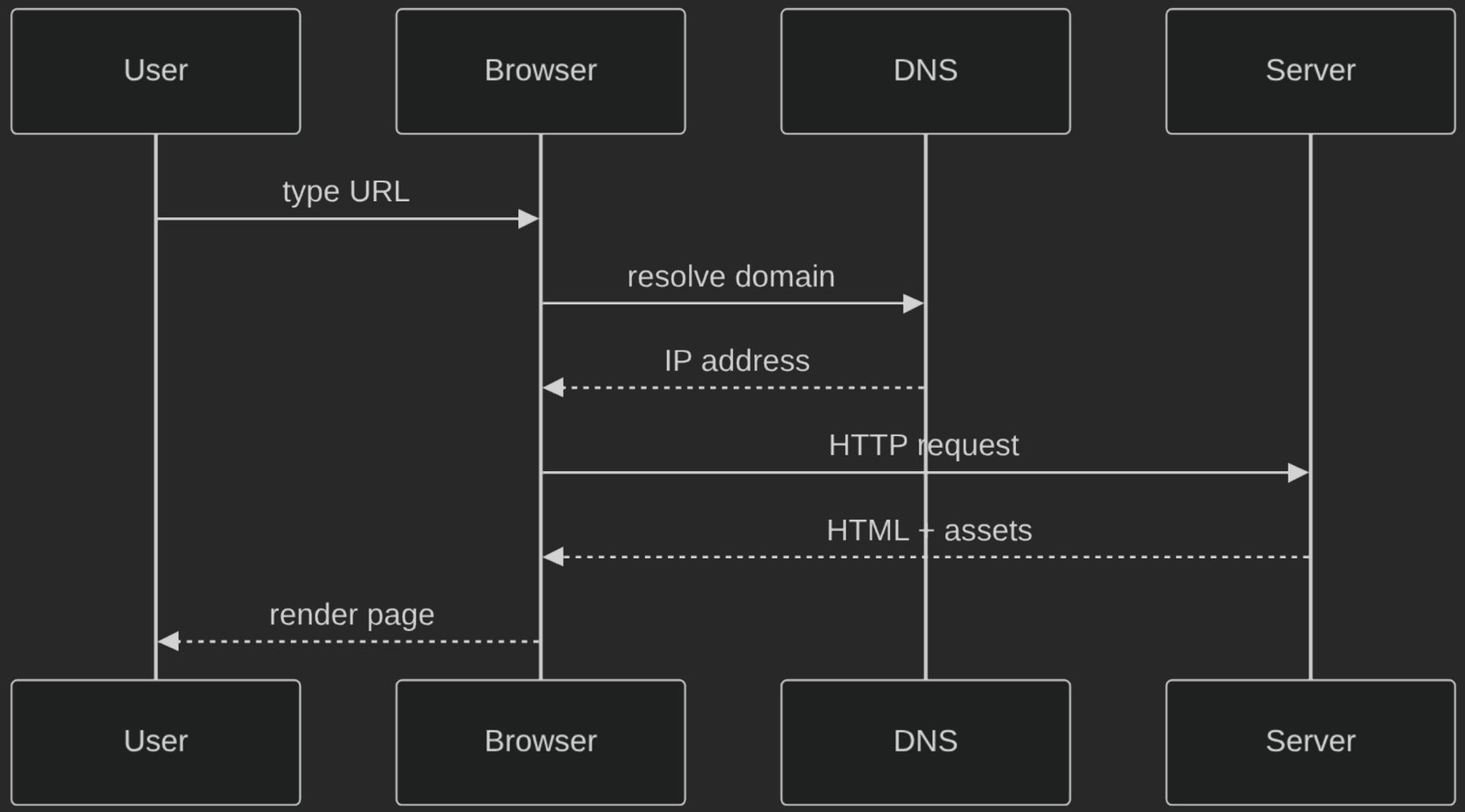
How the Web Works

Step-by-step workflow from browser to server:

1. User enters https://example.com
2. Browser resolves domain → IP
3. Browser sends HTTP request
4. Server responds with HTML, CSS, JS
5. Browser renders page



Section 1 — Web Pages (HTML, CSS, JS)

In this first part we focus on **how a page is built and served**. Do not worry about APIs or JSON yet — just think in terms of:

- HTML = structure & content
- CSS = visual style
- JavaScript = interactivity in the browser

Later sections will reuse this when we load data from APIs instead of hard-coding it.

HTML — Building Blocks

HTML describes structure/content of web pages (like blueprints of a house).

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <title>Hello Web</title>
  </head>
  <body>
    <h1>Welcome!</h1>
    <p>This is my first web page.</p>
    <a href="https://python.org">Python.org</a>
  </body>
</html>
```

- Tags usually come in pairs: `<tag>content</tag>`
- Attributes add metadata: `<a href="https://python.org">`
- Browser builds the **DOM tree** from these nested tags

HTML Storyboard: Code → Server → Browser

1. **Write HTML** (index.html)
2. **Serve it** (python3 -m http.server 8000)
3. **View in browser** (http://localhost:8000)

```
[index.html] --(http.server)--> [localhost:8000] --(render)--> [Page]
```

Focus questions:

- Do tags nest correctly? (<main> contains <section>)
- Did you include alt text for images?
- Can you describe each section aloud?

HTML Document Anatomy

Section	Purpose
<!DOCTYPE>	Tells browser to use modern HTML5 rules
<html>	Root element wrapping everything
<head>	Metadata, title, linked CSS/JS
<body>	Visible content: text, images, forms

Example with more tags:

```html
<body>
  <header><h1>Cyber Café</h1></header>
  <main>
    <section>
      <h2>Menu</h2>
      <ul>
        <li><strong>Espresso</strong> — 3 AZN</li>
        <li><strong>Tea</strong> — 2 AZN</li>
      </ul>
    </section>
    <img src="assets/cafe.jpg" alt="Cafe interior" />
  </main>
  <footer>&copy; 2025 Cyber Café</footer>
</body>
```

Core HTML Tags & Patterns

Tag	What it does	Example snippet
`<h1>`	Page title	`<h1>News</h1>`
`<p>`	Paragraph	`<p>Breaking updates...</p>`
`<a>`	Link to another page/file	`<a href="/about.html">About</a>`
`<img>`	Display image	`<img src="logo.png" alt="Logo" />`
`<ul>/<ol>`	Lists (unordered/ordered)	`<ul><li>Item</li></ul>`
`<form>`	User input	`<form><input name="email" /></form>`
`<div>/<span>`	Layout/grouping	`<div class="card">...</div>`

Tip: Start simple, add CSS later to style it.

Styling with CSS

CSS (Cascading Style Sheets) controls presentation.

Ways to include CSS:

- Inline: `<p style="color:red">`
- `<style>` block in `<head>`
- External file: `<link rel="stylesheet" href="styles.css" />`

Example:

```html
<head>
  <link rel="stylesheet" href="assets/styles.css" />
</head>
<body>
  <h1 class="hero">Welcome</h1>
</body>
```

```css
.hero {
  color: #ff9900;
  text-align: center;
  border-bottom: 2px solid #444;
}
```

Guideline: keep structure (HTML) separate from style (CSS) for clarity.

JavaScript — Introduction

JavaScript is a **programming language** that runs in web browsers (and servers).

Key characteristics:

- **Dynamic** — variables can change types
- **Interpreted** — runs directly without compilation
- **Event-driven** — responds to user actions (clicks, typing, etc.)
- **Versatile** — used for both frontend (browser) and backend (Node.js)

Why JavaScript matters:

- Makes web pages **interactive**
- Can update page content without reloading
- Communicates with servers (APIs)
- Essential for modern web development

Think of it as the "brain" that makes static HTML pages come alive.

JavaScript Basics — Variables & Types

JavaScript has similar concepts to Python, with different syntax.

Variables:

```javascript
let name = "Alice";        // can be reassigned
const age = 25;            // constant, cannot be reassigned
var oldStyle = "avoid";     // old way, avoid in modern code
```

Data Types:

```javascript
let text = "Hello";         // string
let number = 42;            // number
let isActive = true;        // boolean
let data = null;            // null (no value)
let items = [1, 2, 3];      // array (like Python list)
let user = {                // object (like Python dict)
    name: "Alice",
    age: 25
};
```

Functions:

```javascript
function greet(name) {
    return "Hello, " + name;
}

// Arrow function (modern style)
const greet2 = (name) => {
    return "Hello, " + name;
};
```

JavaScript vs Node.js

JavaScript (Browser)

- Runs in the **browser**
- Can manipulate HTML/CSS
- Access to document, window objects
- Cannot access file system (security)
- Used for: interactive web pages, form validation, animations

Example:

```
// Runs in browser
document.getElementById("btn").onclick = function() {
    alert("Clicked!");
};
```

Node.js (Server)

- JavaScript runtime for **servers**
- Built on Chrome's V8 engine
- Can access file system, databases
- No document or window objects
- Used for: web servers, APIs, command-line tools

Example:

```
// Runs on server (Node.js)
const fs = require('fs');
fs.readFile('data.txt', 'utf8', (err, data) => {
    console.log(data);
});
```

Key Point: Same language, different environments. In this module, we focus on **browser JavaScript**.

JavaScript in HTML

JavaScript brings interactivity to web pages.

Ways to include JavaScript:

1. **Inline** (simple actions):

```
<button onclick="alert('Hello!')">Click me</button>
```

2. **Script tag** (inside HTML):

```
<script>
    console.log("Page loaded!");
</script>
```

3. **External file** (recommended):

```
<script src="app.js"></script>
```

Simple example:

```
<body>
  <p id="message">Click the button</p>
  <button onclick="changeText()">Click me</button>

  <script>
    function changeText() {
      document.getElementById("message").textContent = "Hello from JavaScript!";
    }
  </script>
</body>
```

Place <script> tags near the end of <body> so HTML loads first.

JavaScript — Practical Example

Here's a more complete example that demonstrates common patterns:

```html
<body>
  <h1>Counter App</h1>
  <p>Count: <span id="count">0</span></p>
  <button id="increment">+</button>
  <button id="decrement">-</button>
  <button id="reset">Reset</button>

  <script>
    let count = 0;
    const countDisplay = document.getElementById("count");
    const incrementBtn = document.getElementById("increment");
    const decrementBtn = document.getElementById("decrement");
    const resetBtn = document.getElementById("reset");

    function updateDisplay() {
      countDisplay.textContent = count;
    }

    incrementBtn.addEventListener("click", () => {
      count++;
      updateDisplay();
    });

    decrementBtn.addEventListener("click", () => {
      count--;
      updateDisplay();
    });

    resetBtn.addEventListener("click", () => {
      count = 0;
      updateDisplay();
    });
  </script>
</body>
```

What's happening:

1. **Variables** (`count`) store state
2. `getElementById()` finds HTML elements
3. `addEventListener()` responds to clicks
4. `textContent` updates what's displayed
5. **Functions** organize the logic

This pattern (select element → listen for events → update content) is the foundation of interactive web pages.

Serving index.html Quickly

Browsers look for index.html by default.

```
cd website_project
python3 -m http.server 8000
```

- Place index.html alongside the command
- Visit http://localhost:8000 to see it rendered
- Great for checking static prototypes before moving to frameworks

Challenge: create index.html with header, paragraph, list, and image, then serve it locally.

Section 2 — HTTP & APIs

Now that you know what a **page** looks like, we zoom in on the **conversation** between browser and server.

Questions to keep in mind:

- Who is the **client** in this example? Who is the **server**?
- What is the **URL**, what is the **method** (GET/POST/...), and what is inside the **body** (if any)?
- Is the response returning **HTML** or **JSON**?

Once this feels comfortable, we will replace “browser” with **Python code** using requests.

HTTP — The Conversation Rules

HTTP (HyperText Transfer Protocol) defines how clients & servers talk — like a script both sides follow.

Structure:

```
<METHOD> <PATH> HTTP/1.1
Header: Value
Header: Value

<optional body>
```

Request example:

```
GET /api/users HTTP/1.1
Host: example.com
User-Agent: Mozilla/5.0
Accept: application/json
```

Response example:

```
HTTP/1.1 200 OK
Content-Type: application/json

{"status": "ok", "count": 2}
```

Common verbs:

- GET (read), POST (create), PUT/PATCH (update), DELETE (remove)

HTTP Anatomy — Request vs Response

Request

```
POST /login HTTP/1.1
Host: mysite.com
Content-Type: application/json
Authorization: Bearer <token>

{"email": "user@example.com", "password": "***"}
```

- Method + path
- Headers describe metadata (auth, format)
- Body carries data (optional)

Response

```
HTTP/1.1 200 OK
Content-Type: application/json
Set-Cookie: session=abc123

{"status": "ok", "message": "Welcome"}
```

- Status line communicates result
- Headers for caching, cookies, etc.
- Body returns JSON/HTML/binary

Status Codes

Code	Meaning	Use case
200	OK	Successful request
201	Created	Resource created (POST)
301	Moved Permanently	Redirect
400	Bad Request	Invalid client data
401	Unauthorized	Missing/invalid auth
404	Not Found	Resource missing
500	Internal Server Error	Server bug/problem

Fun reference with cat photos: https://http.cat

APIs — Digital Waiters

API = Application Programming Interface.

- Think of a restaurant: you (client) tell the waiter (API) what you want, they talk to the kitchen (server), and bring back the dish (data).
- HTTP APIs expose data/actions via URLs, verbs, and JSON payloads.
- Good APIs document their menu (endpoints, parameters, status codes).

Example: Tasks API served from http://localhost:3000

Endpoint	Method	Description
/api/tasks?completed=true	GET	List tasks filtered by completion flag
/api/tasks	POST	Create a new task with title/description flags
/api/tasks/{id}	GET	Fetch one task by its numeric identifier
/api/tasks/{id}	PUT	Update title, description, or completed status
/api/tasks/{id}	DELETE	Remove the task forever
/api/health	GET	Quick health check for the service

Menu analogy:

- Endpoint = dish name
- Query params/body = customization
- Response JSON = meal you receive

JSON — Data Containers

JSON (JavaScript Object Notation) represents structured data with nested objects/lists.

```
{
  "id": 42,
  "username": "admin",
  "scores": [99, 88, 91],
  "profile": {
    "city": "Baku",
    "active": true
  }
}
```

Analogy: dictionaries + lists combined.

Rules of thumb:

- Keys in double quotes, values can be string/number/bool/null/object/array
- Order usually not important, whitespace ignored

Another sample (array of objects):

```
[
  {"task": "learn HTML", "done": true},
  {"task": "build API client", "done": false}
]
```

Analogy: shipping boxes inside a truck — arrays hold boxes, each box (object) has labeled compartments (keys).

From HTTP & JSON to Python Code

Let’s connect the dots:

1. Browser or client sends an **HTTP request** (method + path + headers + body).
2. Server sends back an **HTTP response** with a **status code** and **body**.
3. If `Content-Type: application/json`, the body contains **JSON**, which feels like Python `dict` + `list`.
4. Our Python script with `requests` plays the role of the **client** instead of the browser.

So when you see `response.json()` later, imagine:

> “Take the JSON body from the HTTP response and turn it into Python dictionaries/lists I can work with.”

Inspecting APIs

Tools:

- Browser dev tools → Network tab
- `curl` in terminal:

```
curl -G "http://localhost:3000/api/tasks" \
  --data-urlencode "completed=true"
```

- Postman, Bruno, or Insomnia for graphical testing

Always check: URL, method, headers, body, response code, payload. Tasks API also requires you to pass `completed` as a query parameter when listing.

Python requests Basics

```python
import requests

url = "http://localhost:3000/api/tasks"
params = {"completed": "false"}  # API requires this flag
response = requests.get(url, params=params, timeout=5)

print("Status:", response.status_code)
response.raise_for_status()
tasks = response.json()
print("First task title:", tasks[0]["title"] if tasks else "No tasks yet")
```

snippet +exec is disabled, run with -x to enable

Notes:

- `timeout` prevents hanging forever (network problems)
- `response.status_code` shows the HTTP status (200, 404, 500, …)
- `response.raise_for_status()` throws an exception for 4xx/5xx errors
- `response.text` is the raw string body, `response.json()` parses JSON into Python types

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Checkpoint — Your First API Call

1. Activate your venv
2. pip install requests
3. Run:

```python
import requests

BASE_URL = "http://localhost:3000/api/tasks"
data = requests.get(
    BASE_URL,
    params={"completed": "true"},
    timeout=5,
).json()

print("Total completed tasks:", len(data))
print("Sample record:", data[0] if data else "None")
```

4. Screenshot output for homework

Questions to consider:

- What happens if you forget .json()?
- How would you handle an empty list or missing key?

Section 3 — Python as Web Client & Simple Server

In this final part we:

1. Use Python + requests as an **HTTP client** talking to the Tasks API.
2. Use http.server (and a tiny custom server) as a **simple HTTP server** to serve static HTML.

Key mental model:

- Browser and Python scripts are just **different HTTP clients**.
- Servers (your simple Python server, or the Tasks API on port 3000) all speak the same HTTP “language”.

When in doubt, go back to the HTTP request/response examples and map each line to what requests or the browser is doing.

Installing requests Inside a venv

```python3 -m venv .venv
source .venv/bin/activate  # Windows: .venv\Scripts\activate
pip install --upgrade pip
pip install requests
pip freeze
```

- Always isolate dependencies per project
- `pip freeze` helps document versions (`requirements.txt`)
- Deactivate with `deactivate` when done

Sending Data with requests

```python
import requests

payload = {
    "title": "Draft Module 8 slides",
    "description": "Cover Tasks API examples",
    "completed": False,
}

resp = requests.post(
    "http://localhost:3000/api/tasks",
    json=payload,
    timeout=5,
)

resp.raise_for_status()
created = resp.json()
print("New task id:", created["id"])
print("Completed flag:", created["completed"])
```

snippet +exec is disabled, run with -x to enable

`json=...` auto converts dict to JSON & sets `Content-Type: application/json`.

Serving Static HTML with Python

Quick demo using built-in http.server.

```
python3 -m http.server 8000
```

- Serves current directory
- Visit http://localhost:8000 to view files
- Great for testing HTML/CSS locally

Stop with Ctrl+C.

Custom Minimal Server

Serve a specific HTML file with Python.

```python
from http.server import SimpleHTTPRequestHandler, HTTPServer
from pathlib import Path

PORT = 8080
ROOT = Path("website")

class StaticHandler(SimpleHTTPRequestHandler):
    def translate_path(self, path):
        # Map requests into our ROOT directory
        return str(ROOT / path.lstrip("/"))

if __name__ == "__main__":
    ROOT.mkdir(exist_ok=True)
    httpd = HTTPServer(("0.0.0.0", PORT), StaticHandler)
    print(f"Serving on http://localhost:{PORT}")
    httpd.serve_forever()
```

snippet +exec is disabled, run with -x to enable

Place `index.html` inside `website/` and run script.

Explanation:

- `HTTPServer` listens on port 8080 and waits for requests
- `SimpleHTTPRequestHandler` knows how to serve files; we override `translate_path` to force it into `website/`
- `ROOT.mkdir(exist_ok=True)` ensures the folder exists before serving
- Visit http://localhost:8080/index.html to test

Project structure reference:

```
website/
├── index.html
├── about.html
└── assets/
    └── styles.css
server.py
```

Run `python server.py` from the folder containing `website/`.

Meme Slide

Thanks!

By ElnurBDa